

CoRoT-2b

R-1439

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_140729 · created 2026-08-01T08:13:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456867.7143 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.018
Transit depth (Rp/Rs)^2	2.39 +/- 0.54 [%]
Orbital Inclination (inc)	83.62 +/- 0.93
Airmass coefficient 1 (a1)	1.259 +/- 0.019
Airmass coefficient 2 (a2)	-0.193 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	3.35
Optimal Annulus	8.61
Transit Duration (day)	0.0727 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.018 vs 0.1667 (deviation 0.65σ)
Duration fit vs published	0.0727 d vs 0.095 d (deviation 2.81σ)
Mid-time O–C	-6.7 min
Depth SNR	8.6
Residual scatter	1.53 %

Light curve

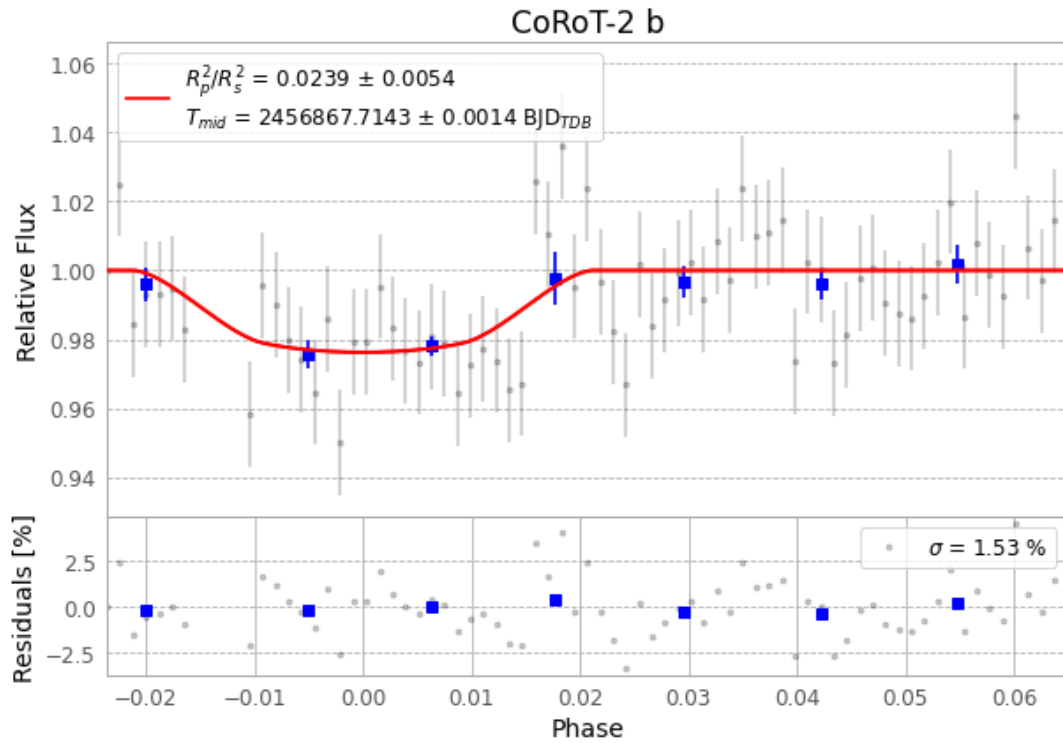


Figure 1 — FinalLightCurve_CoRoT-2 b_2014-07-28.png (SHA-256 3bb98b1f92cd9027...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7ab2cabccc710e13...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), CoRoT-2140729040012.FITS → CoRoT-2140729074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-140729.log (8abac834773a...), FinalLightCurve_CoRoT-2 b_2014-07-28.png (3bb98b1f92cd...), FinalLightCurve_CoRoT-2 b_2014-07-28.csv (1d084106a9e0...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 08:13Z.